Cheat Codes

PHYSICS

1.) Logic Gate Boolean Outputs [2025, 2024, 2023, 2022, 2021, 2020]

OR : $Y = A + B$

AND : $Y = A \cdot B$

NAND : $Y = \overline{A \cdot B}$

2.) de-Broglie wavelength [2025, 2024, 2022, 2021, 2020]

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$$

3.) Bohr's Model (Energy & Radius) [2025, 2024, 2023, 2020, 2019]

$$E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$$

$$r_n = 0.529 \frac{n^2}{Z} \text{ Å}$$

4.) Ideal Gas Eqn & variations [2025, 2024, 2023, 2021, 2020, 2019]

$$PV = nRT$$

$$\rho = \frac{PM}{RT}$$

5.) Eqn of Motion (Constn Accn) [2025, 2023, 2022, 2021, 2020]

$$v^2 - u^2 = 2aS$$

$$S = ut + \frac{1}{2}at^2$$

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6.) Capacitance (Parallel plate & Energy) [2025, 2024, 2023, 2021, 2020, 2019]

$$C = \frac{k\epsilon_0 A}{d} \qquad U = \frac{1}{2}CV^2$$

7.) Magnetic Moment [2025, 2024, 2022, 2021]

$$M = NIA$$

8.) Distance in n^{th} second [2025, 2022, 2021]

$$S_n = u + \frac{a}{2}(2n-1)$$

9.) Power of Lens & Combinations [2025, 2023, 2021, 2024]

$$P = \frac{1}{f} \qquad P_{eq} = P_1 + P_2 + \dots$$

10.) Magnetic force on a Moving Charge [2025, 2023, 2021]

$$F = qvB\sin\theta$$

11.) Radioactive Decay Activity [2025, 2023, 2021]

$$A = \frac{A_0}{2^n} \qquad n = \frac{t}{T_{1/2}}$$

12.) Critical Angle & Total Internal Reflection [2025, 2023, 2022]

$$\sin \theta_c = \frac{1}{\mu}$$

13.) Young's Double Slit Exp: (fringe width) [2022, 2024, 2020, 2019]

$$\beta = \frac{\lambda D}{d}$$

14.) Dimensional Analysis for Constants [2022, 2021, 2020]

$$[G] = [M^{-1} L^{3} T^{-2}]$$

$$[\text{Stress}] = [M L^{-1} T^{-2}]$$

15.) Electric potential (Point Charge/sphere) [2022, 2021, 2020]

$$V = \frac{1}{4\pi \epsilon_0} \frac{Q}{R}$$

16.) Avg. speed (Equal distances travelled) [2025, 2023]

$$V_{avg} = \frac{2 v_1 v_2}{v_1 + v_2}$$

17.) Escape velocity [2023, 2021]

$$v_e = \sqrt{\frac{2GM}{R}}$$

18.) Terminal Voltage of a Cell [2025, 2024]

$$V = E - Ir$$

19.) Organ pipe frequencies (fundamental) [2025, 2023]

$$f_{open} = \frac{v}{2L}$$

$$f_{closed} = \frac{v}{4L}$$

20.) Kepler's 3rd law [2025, 2023]

$$T^2 \propto R^3$$

$$T = 2\pi\sqrt{\frac{R^3}{GM}}$$

21.) Lens Maker formulae [2023, 2022]

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

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22.) RMS speed of Gas Molecule [2025, 2023]

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

23.) Mechanical Power [2022, 2021]

$$P = \vec{F}.\vec{v}$$
$$= Fv\cos\theta$$

24.) Simple Pendulum time period [2024, 2021]

$$T = 2\pi\sqrt{\frac{l}{g}}$$

25.) SHM Poth energy freq: [2023, 2021]

$$f_{PE} = 2 \times f_{oscillations}$$

26.) Ampere's law (st. wire) [2025, 2023, 2019]

$$B = \frac{\mu_0 I}{2\pi r}$$

27.) Error analysis (Multiplication of Power) [2025, 2023, 2019]

$$\frac{\Delta Z}{Z} = a\frac{\Delta A}{A} + b\frac{\Delta B}{B}$$

28.) Minimum wavelength of X-rays [2023]

$$\lambda_{min} = \frac{hc}{eV}$$

29.) RMS Voltage (AC) [2024, 2022]

$$V_{rms} = \frac{V_0}{\sqrt{2}}$$

30.) Grav. Potⁿ energy [2019, 2021, 2024]

$$U = -\frac{GmM}{R}$$

31.) Centre of Mass (2 particles) [2020, 2022]

$$x_{com} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

32.) Current density & Ohm's law : [2021, 2022]

$$J = \sigma E$$

$$J = \frac{I}{A}$$

33.) Brewster law : $\mu = \tan i_p$
[2020, 2025]

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34.) Rᵗʰ of Combined liqᵈ drops [2021]

$$V_{Big} = n^{2/3} V_{small}$$

35.) Accⁿ in SHM [2023, 2020]

$$a = -\omega^2 x$$

36.) Impulse - momentum theorem [2025, 2023]

$$I = \Delta p = m(v_2 - v_1)$$

37.) Magnetic field at centre of an Arc. [2023, 2022]

$$B = \frac{\mu_0 I}{4\pi R}\theta$$

38.) Work done by variable force [2019]

$$W = \int F.dx$$

39.) Beat freq [2020]

$$f_{beats} = |f_1 - f_2|$$

40.) Nucleus radius [2022]

$$R = R_0 A^{1/3}$$

—— x —— x ——